Unit: Unit 8: Quadratics
Topic: Solving by Factoring and Other Methods

Name: ______________
Date: __________

Advanced (Habanero)

Q1. Solve $x^2 + 5x + 6 = 0$ by factoring.
(A) $x = -2, x = -3$
(B) $x = -1, x = -6$
(C) $x = 2, x = 3$
(D) $x = 1, x = 6$
(E) $x = -2, x = 3$

Q2. Solve $x^2 - 7x + 12 = 0$ by factoring.
(A) $x = 3, x = 4$
(B) $x = 5, x = 12$
(C) $x = 3, x = 12$
(D) $x = 4, x = 12$
(E) $x = 5, x = 3$

Q3. Verify the solution to $x^2 + 2x - 6 = 0$ is $x = -3, x = 2$.
(A) True
(B) False
(C) Maybe
(D) Yes
(E) No

Q4. Solve $x^2 + 2x - 15 = 0$ by factoring.
(A) $x = -5, x = 3$
(B) $x = 5, x = 3$
(C) $x = -5, x = -3$
(D) $x = 3, x = 5$
(E) $x = 5, x = -3$

Q5. Solve $x^2 - 9x + 20 = 0$ by factoring.
(A) $x = 4, x = 5$
(B) $x = 5, x = 4$
(C) $x = 4, x = 9$
(D) $x = 5, x = 9$
(E) $x = 4, x = 1$

Q6. Verify the solution to $x^2 - 4x - 5 = 0$ is $x = -1, x = 5$.
(A) True
(B) False
(C) Maybe
(D) Yes
(E) No

Q7. Solve $x^2 + x - 12 = 0$ by factoring.
(A) $x = -4, x = 3$
(B) $x = -3, x = 4$
(C) $x = 4, x = 3$
(D) $x = -4, x = -3$
(E) $x = -3, x = 3$

Q8. Solve $x^2 - 2x - 15 = 0$ by factoring.
(A) $x = -3, x = 5$
(B) $x = 3, x = 5$
(C) $x = 5, x = -3$
(D) $x = -5, x = 3$
(E) $x = 5, x = 3$

Open Ended Questions (FRQ)

Q9. Solve $x^2 + 4x + 4 = 0$ and verify your solution.

Q10. Solve $x^2 - 5x - 6 = 0$ by factoring and verify your solution.

Chef's Tips

- Tip 1: Make sure to check your work.
- Tip 2: Use the correct formula for factoring.
- Tip 3: Read the questions carefully.